# CYCLISTIC BIKE-SHARE CASE STUDY

*How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?*

Google Data Analytics Professional Certificate — Capstone Case Study

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Tools Used: Python (Jupyter Notebook) | Tableau Public | Google Sheets

## 1. ASK

### Business Task

Cyclistic, a Chicago-based bike-share company, wants to maximize the number of annual memberships. The director of marketing, Lily Moreno, believes converting casual riders into annual members is key to future growth. As a junior data analyst, I was assigned to answer the following business question:

***"How do annual members and casual riders use Cyclistic bikes differently?"***

### Key Stakeholders

- Lily Moreno — Director of Marketing and manager
- Cyclistic Executive Team — Will approve or reject marketing recommendations
- Cyclistic Marketing Analytics Team — Collaborators on data-driven strategy

### Guiding Questions

- What is the problem we are trying to solve?
- How can our insights drive business decisions?

# 2. PREPARE

## Data Sources Used

Data was sourced from Divvy Bikes historical trip data, made available by Motivate International Inc. under a public license. Two quarterly datasets were used for this analysis:

- Divvy_Trips_2019_Q1.csv — 365,069 rows, 12 columns
- Divvy_Trips_2020_Q1.csv — 426,887 rows, 13 columns
- Combined dataset: 784,285 total rides after merging and cleaning

## Data Credibility — ROCCC Assessment

| Criteria | Assessment | Details |
|---|---|---|
| Reliable | Yes | Data collected directly from Divvy bike-share system |
| Original | Yes | First-party data from Motivate International Inc. |
| Comprehensive | Partial | Q1 only (Jan-Mar 2019 & 2020) — 3 months of data |
| Current | Partial | 2019-2020 data — not the most recent |
| Cited | Yes | Publicly available under Motivate International license |

## Data Privacy

Personally identifiable information (PII) was not used. Rider names, credit card numbers, and personal details are excluded from the dataset in compliance with data privacy regulations.

## Data Limitations

- Only Q1 data (January, February, March) — seasonal bias toward winter months
- No demographic data available for deeper segmentation
- Cannot link individual casual riders across multiple trips

# 3. PROCESS

## Tools Used

- Python (Jupyter Notebook) — Data cleaning, merging, and transformation
- pandas library — Data manipulation
- Google Sheets — Data verification and pivot table validation
- Tableau Public — Data visualization and dashboard creation

## Data Cleaning Steps

The following steps were performed in Python to clean and prepare the data:

- Imported both CSV files using pandas read_csv()
- Renamed columns in 2019 dataset to match 2020 column names (trip_id → ride_id, start_time → started_at, usertype → member_casual)
- Merged both datasets using pd.concat() — resulting in 784,285 rows
- Standardized member_casual column: renamed 'Subscriber' → 'member' and 'Customer' → 'casual'
- Created ride_length column: calculated duration using ended_at minus started_at
- Created day_of_week column using dt.dayofweek (0=Monday, 6=Sunday)
- Created month column using dt.month
- Removed rides with zero or negative duration (invalid entries)
- Removed rides shorter than 1 minute (test/maintenance rides)
- Verified no duplicate ride_id values — 0 duplicates found
- Created ride_length_minutes column for numeric analysis

## Data Verification

The cleaned dataset was exported as final_cyclistic1_summary.csv and verified in Google Sheets:

- Google Sheets showed 784,286 rows including header — confirming 784,285 rides
- COUNTA pivot table confirmed: member = 716,593 rides, casual = 67,692 rides
- Total verified: 716,593 + 67,692 = 784,285 ✓

# 4. ANALYZE

## Summary Statistics

| Metric | Casual Riders | Annual Members |
|---|---|---|
| Total Rides | 67,692 | 716,593 |
| Percentage of Total | 8.6% | 91.4% |
| Average Ride Duration | 89.79 minutes | 13.32 minutes |
| Peak Day | Saturday & Sunday | Tuesday & Wednesday |
| Peak Month | March | March |

## Key Findings

**Finding 1 — Ride Frequency:** Annual members take significantly more rides than casual riders. Members account for 91.4% of total rides (716,593) compared to only 8.6% for casual riders (67,692). This indicates members use the service regularly for daily needs.

**Finding 2 — Ride Duration:** Casual riders average 89.79 minutes per ride compared to 13.32 minutes for members. Casual riders ride nearly 7 times longer per trip, suggesting leisure-oriented usage rather than commuting.

**Finding 3 — Day of Week Pattern:** Members show consistent usage across weekdays (Monday-Friday), peaking on Tuesday and Wednesday. Casual riders show significantly higher usage on Saturday and Sunday, indicating recreational and leisure riding behavior.

**Finding 4 — Monthly Trend:** Both groups show an upward trend from January to March, with March having the highest ridership. This is consistent with improving weather conditions in Chicago as spring approaches.

# 5. SHARE

## Visualizations Created

All visualizations were created in Tableau Public and published at:

https://public.tableau.com/app/profile/shifana.thasni/viz/CyclisticBikeShareAnalysis

### Chart 1 — Number of Rides by Member Type

A bar chart comparing total rides between casual riders (67,692) and annual members (716,593). This chart clearly shows that members use the service at a significantly higher rate, establishing the baseline difference between the two user groups.

### Chart 2 — Number of Rides by Day of Week

A stacked bar chart showing ride counts for each day of the week, split by member type. Key insight: members dominate on weekdays while casual riders show relative increases on weekends, revealing fundamentally different usage patterns.

### Chart 3 — Number of Rides by Month (Q1)

A line chart showing the monthly trend for January, February, and March. Both user groups show increasing ridership toward March. The member line is significantly higher throughout, but both show the same upward seasonal trend.

### Chart 4 — Average Ride Duration by Member Type

A bar chart comparing average ride length. Casual riders average 89.79 minutes versus 13.32 minutes for members. This is the most powerful visualization — it demonstrates that casual riders are highly engaged cyclists who spend more time on each ride, making them strong candidates for membership conversion.

### Dashboard

All four charts were combined into a single professional dashboard titled 'Cyclistic Bike Share Analysis' and published to Tableau Public for executive review.

# 6. ACT

## Conclusions

Based on the analysis of 784,285 Cycle rides from Q1 2019 and Q1 2020, the data clearly shows that annual members and casual riders use bikes in fundamentally different ways:

- Members are daily commuters who take short, frequent trips on weekdays
- Casual riders are leisure cyclists who take longer trips on weekends
- Casual riders are already engaged with the platform — they ride 7x longer per trip
- Both groups increase usage in spring, suggesting seasonal marketing opportunities

## Top 3 Recommendations

### Recommendation 1 — Launch a Weekend or Leisure Membership Plan

Since Casual riders peak on weekends for leisure. Offering a weekend-only annual membership at a reduced price point would directly address their usage pattern. This removes the barrier of paying for a full annual membership when they only ride recreationally. Digital advertising on social media platforms targeting Chicago weekend activities could effectively reach this audience.

### Recommendation 2 — Spring Conversion Campaign (February-March)

Both casual riders and members show a significant increase in ridership in March. This seasonal uptick represents the best window to target casual riders with membership conversion offers. A limited-time spring promotion offering discounted annual membership rates in February and March — when casual riders are increasing their usage — would maximize conversion likelihood. Email campaigns, in-app notifications, and station-based advertising at high-casual-traffic locations would be effective channels.

### Recommendation 3 — Highlight Cost Savings Through Digital Media

Casual riders average 89.79 minutes per ride. At single-ride or day-pass rates, frequent casual riders are likely spending significantly more than the annual membership cost. A digital media campaign showing a personalized cost comparison — 'Based on your riding habits, you could save $X with an annual membership' — would create a compelling financial incentive. This can be delivered through the Cyclist app, targeted social media ads, and email campaigns to registered casual riders.

Business Impact

The insights from this analysis can help to improve its membership conversion strategy by targeting casual riders based on their riding behaviour. Since casual riders are highly active on weekends and spend significantly longer on each trip, introducing flexible membership plans and seasonal promotions could increase conversion rates.

implementing these recommendations may achieve:

• Increased annual membership subscriptions, leading to higher recurring revenue

• Improved customer retention through personalized membership options

• More effective marketing campaigns by targeting high-engagement casual riders

• Better resource allocation by focusing on peak usage periods and high-traffic stations

• Stronger long-term business growth through a more stable customer base

## Additional Data That Would Strengthen Analysis

- Full year data (Q2, Q3, Q4) — to identify seasonal patterns across all 12 months
- Geographic data — to identify which stations have highest casual rider traffic for targeted marketing
- Pricing data — to calculate actual cost savings for individual casual riders
- Return visitor data — to identify casual riders who ride frequently enough to benefit from membership

## Next Steps

- Present dashboard and recommendations to Lily Moreno and executive team
- Marketing team to design spring conversion campaign for February-March launch
- Product team to evaluate feasibility of weekend membership pricing tier
- Digital team to develop personalized cost-savings calculator for casual riders
- Collect full-year data to validate seasonal patterns and refine targeting

# APPENDIX

## Tools and Technologies

| Tool | Purpose | Output |
|---|---|---|
| Python (Jupyter Notebook) | Data cleaning and transformation | Cleaned dataset — 784,285 rows |
| pandas library | Data manipulation and groupby analysis | Summary CSV file |
| Google Sheets | Data verification | Pivot table validation |
| Tableau Public | Data visualization | 4 charts + dashboard |
| GitHub / Portfolio | Project documentation | Public portfolio link |

## Data Cleaning Summary

| Step | Action | Result |
|---|---|---|
| 1 | Merged 2019 Q1 and 2020 Q1 datasets | 791,956 combined rows |
| 2 | Renamed columns for consistency | Standardized column names |
| 3 | Renamed Subscriber/Customer labels | member/casual only |
| 4 | Removed zero/negative duration rides | Invalid rides removed |
| 5 | Removed rides under 1 minute | Test rides removed |
| 6 | Verified no duplicates | 0 duplicates found |
| Final | Clean dataset | 784,285 rows, 9 columns |

## References


- Divvy Bikes Trip Data — Motivate International Inc. (https://divvybikes.com/system-data)
- Google Data Analytics Professional Certificate — Coursera
- Tableau Public Dashboard — https://public.tableau.com/app/profile/shifana.thasni